T20 World Cup 2022 Cricket Analytics

Final Analysis Report By Sourav Roychoudhury

Tools: Python | Pandas | Power BI | Power Query | DAX | Bright Data

Executive Summary

The T20 World Cup 2022 Cricket Analytics Project is an end-to-end sports analytics solution that leverages web scraping, Python, Pandas, Power BI, Power Query, and DAX to evaluate player performances throughout the ICC Men's T20 World Cup 2022. The primary objective is to identify the Best Playing XI based entirely on statistical performance rather than reputation or subjective judgment.

Using match data collected from ESPN Cricinfo through Bright Data, the project analyzes batting, bowling, and all-round performance using role-specific metrics. Interactive Power BI dashboards enable dynamic filtering and comparison of players across multiple performance indicators such as strike rate, batting average, economy rate, boundary percentage, dot-ball percentage, wickets taken, and consistency.

1. Project Objective

The objective of this project is to build an analytical framework capable of selecting the strongest possible T20 cricket team using real match statistics from the ICC Men's T20 World Cup 2022. The project aims to:

- Collect and consolidate tournament data from multiple matches
- Clean and transform raw JSON data into structured datasets
- Analyze player performance across different roles
- Develop interactive Power BI dashboards for visual exploration
- Select the Best XI using predefined statistical criteria

Central business question: Which eleven players performed well enough statistically to form the strongest possible T20 team during the tournament?

2. Dataset Overview

The dataset was collected by scraping ESPN Cricinfo using Bright Data and includes match summaries, player scorecards, batting statistics, bowling statistics, match results, and player profiles.

Batting Metrics

Metric	Metric	Metric	Metric
Runs	Balls Faced	Batting Average	Strike Rate
Boundary Percentage	Batting Position	Innings Played	Not Outs

Bowling Metrics

Metric	Metric	Metric
Overs Bowled	Runs Conceded	Economy Rate
Wickets	Dot Balls	Bowling Average

Match Information

Field	Field	Field	Field
Opposition	Match Result	Venue	Stage of Tournament

3. Data Collection

The first phase involved automated web scraping using Bright Data. The scraper extracted structured JSON data from ESPN Cricinfo including match summaries, full batting scorecards, bowling scorecards, and player information. The raw JSON files were then stored for preprocessing.

4. Data Cleaning & Transformation

Python and Pandas were used extensively to prepare the dataset for analysis.

Data Cleaning

- Removed duplicate records
- Handled missing values
- Standardized player names
- Corrected inconsistent formats

Feature Engineering

Additional performance metrics were calculated, including:

- Batting Average and Strike Rate
- Boundary Percentage
- Bowling Economy and Dot Ball Percentage

- Bowling Strike Rate

Data Integration

Multiple datasets were merged into a unified analytical model containing batting, bowling, player, and match information.

5. Power Query & Data Modeling

Power Query Transformation

After importing the processed CSV files into Power BI, Power Query was used to rename columns, remove unnecessary fields, merge lookup tables, create relationships, and improve data consistency.

Data Modeling

A star schema was implemented inside Power BI connecting four tables:

Table	Description
Match Table	Match-level details including venue, opposition, and result
Batting Table	Player batting statistics per match
Bowling Table	Player bowling statistics per match
Player Table	Player profiles and role classifications

Relationships between these tables allow dynamic filtering across all dashboard visuals.

6. DAX Measures

Several DAX measures were created to calculate dynamic KPIs:

- Total Runs
- Average Strike Rate
- Batting Average
- Economy Rate
- Total Wickets
- Boundary Percentage
- Dot Ball Percentage

These measures enable interactive filtering without modifying the underlying dataset.

7. Player Role Classification

Players were categorized into five specialized roles, each evaluated using role-specific criteria.

Role	Key Selection Criteria	Primary Metrics
Power Hitters / Openers	High Strike Rate, High Boundary %, Powerplay scoring	Strike Rate, Runs, Boundary %, Balls Faced
Anchors	High Batting Average, Consistent scoring, Partnership building	Average, Runs, Innings, Consistency
Finishers	High Death-Overs Strike Rate, Finishing ability	Strike Rate, Sixes, Boundary %
All-Rounders	Batting + Bowling contribution, Match impact	Runs, Wickets, Economy, Strike Rate
Specialist Bowlers	Economy Rate, Wickets, Dot Ball %, Bowling Strike Rate	Economy, Wickets, Dot Ball %

Each role serves a distinct tactical purpose. Openers maximize powerplay scoring, anchors prevent batting collapses, finishers boost late-innings totals, all-rounders add tactical flexibility, and specialist bowlers control run flow while taking wickets.

8. Dashboard Overview

The Power BI dashboard provides interactive analysis through dedicated pages for each player role. Dashboard features include:

- KPI Cards for key performance indicators
- Player Comparison Tables
- Scatter Plots for multi-metric comparison
- Bar Charts for ranking players
- Performance Filters for threshold-based selection
- Dynamic Role Selection across all pages

Users can filter players based on predefined statistical thresholds to identify the strongest candidates for each role.

9. Performance Analysis

Batting Performance

The dashboard evaluates batting performance using Total Runs, Strike Rate, Batting Average, Boundary Percentage, and Innings Played. Players with both high strike rates and strong batting averages were prioritized to ensure a balance between aggressive scoring and consistency.

Bowling Performance

Bowling effectiveness was measured using Wickets Taken, Economy Rate, Dot Ball Percentage, and Bowling Strike Rate. The analysis favored bowlers capable of maintaining low economy rates while consistently taking wickets, highlighting those who effectively restricted opposition scoring.

All-Round Performance

All-rounders were evaluated using combined batting and bowling statistics. Players who contributed significantly in both disciplines offered greater tactical flexibility and strengthened overall team balance.

Comparative Player Analysis

Interactive visuals allow users to compare multiple players simultaneously across dimensions such as Strike Rate vs Average, Economy vs Wickets, Runs vs Boundary Percentage, and Dot Ball % vs Economy. This enables data-driven player selection without relying solely on total runs or wickets.

10. Final Best XI Selection

The final playing XI was selected after applying role-specific statistical filters across all player categories. Selection criteria included:

- Batting consistency and aggressive scoring ability
- Bowling efficiency and wicket-taking ability
- Match impact and team balance
- Role specialization

The selected XI represents a balanced squad with explosive batting, dependable middle-order stability, versatile all-rounders, and a strong bowling attack.

Key Insights

- Statistical analysis provides a more objective approach to player selection than relying on reputation alone
- High strike rate alone is insufficient; consistency and batting average are equally important
- Specialist bowlers with low economy rates and high dot-ball percentages contribute significantly to match-winning performances
- All-rounders increase tactical flexibility by strengthening both batting depth and bowling options
- Interactive dashboards enable coaches and analysts to explore player performance dynamically across multiple criteria
- Data-driven selection ensures a balanced team composition optimized for different match situations

Business Recommendations

1. **Adopt Data-Driven Selection:** Use objective performance metrics rather than subjective opinions when selecting players for future tournaments.
2. **Define Role-Specific KPIs:** Evaluate players based on metrics relevant to their roles — strike rate for openers, consistency for anchors, economy rate for bowlers.
3. **Monitor Performance Continuously:** Update the dataset after each series or tournament to maintain current player rankings and identify emerging talent.
4. **Expand the Analytics Model:** Include additional metrics such as fielding performance, player fitness, opposition strength, venue conditions, and pressure situations to improve selection accuracy.
5. **Enhance Predictive Capabilities:** Integrate machine learning models to forecast future player performance based on historical trends and match conditions.

Skills Demonstrated

Category	Skills
Data Collection	Web Scraping using Bright Data, JSON parsing
Data Processing	Python, Pandas, Data Cleaning, Feature Engineering
BI & Visualization	Power BI, Power Query, DAX, Dashboard Design
Analytics	Sports Analytics, KPI Development, Trend Analysis
Soft Skills	Analytical Storytelling, Business Intelligence, End-to-End Workflow

Conclusion

The T20 World Cup 2022 Cricket Analytics Project demonstrates how modern data analytics can transform raw sports data into actionable insights for strategic decision-making. By integrating web scraping, Python-based data processing, Power BI data modeling, and interactive dashboard design, the project provides a comprehensive framework for evaluating player performance across multiple dimensions.

The final Best XI is selected through objective, role-specific statistical analysis, illustrating the value of evidence-based decision-making in sports analytics. Beyond cricket, the techniques showcased in this project are applicable to broader domains such as business intelligence, performance optimization, predictive analytics, and operational decision support, making it a strong portfolio project that highlights end-to-end data analytics capabilities.